

Taal

The agent that sells what the forecast says you'll throw away.

TEAM

Nandish · lead

Rinu · co-builder

PROBLEM

- 1 Forecasts and offers live in different tools: one dark store writes stock off while the next runs out.
- 2 Customers get blanket markdowns, not offers matched to what they buy.
- 3 Nothing plans to the online sell-by cut-off: 30% of shelf life or 45 days at delivery.¹

¹ FSSAI advisory to e-commerce FBOs, 3 Dec 2024, file RCD-13/1/2024-Regulatory-FSSAI(E-13150): minimum 30% shelf life or 45 days at delivery. Either-limit reading is Taal's configurable default.

BRIEF ABOUT THE IDEA

One decision loop turns a forecast into a sale, before the write-off happens.

- 1 **Capture** a pallet photo
- 2 **Sense** nightly gaps, in rupees
- 3 **Plan** a play; the estimator owns the rupees
- 4 **Approve** with a holdout; feed it back to the forecast
- 5 **Engage** consenting customers, in their language
- 6 **Measure** treated vs holdout, 95% CI

ONE EXAMPLE · SEEDED DEMO TENANT

6 days

until Masala Chips at dark store DS-07 can no longer be sold online

368
units

₹9,200
at stake

Recorded Gemini run, 28 Sep: rationale rejected 3 times by cite-or-drop, passed on the 4th. Approve: 316 treated, 36 holdout.

DIFFERENTIATION

A holdout on every play. The play written back into the forecast.

	ONLINE SELL-BY CUT-OFF	HOLDOUT PER PROMO	PROMO → FORECAST WRITE-BACK	INDIA / Q-COMMERCE
Taal	Yes	Yes	Yes	Built for it; seeded tenant
Blue Yonder	Unclear	Unclear	Unclear	Partial
RELEX	Partial	Unclear	Partial	Partial
Wasteless	No	Unclear	Unclear	Unclear
Afresh	No	Unclear	Unclear	No
Flashfood	No	No	No	No

Unclear = no public evidence found in vendor pages read on 2026-10-09; it does not mean the feature is absent. Partial and No are our reading of those pages. Afresh and Flashfood are different product categories.

LIST OF FEATURES

Six capabilities, gated by rules a model never touches.



Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.



Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.



Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.



Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.



Customer Agent

Grounded in stock, offers and consent. Orders through an in-process MCP tool.

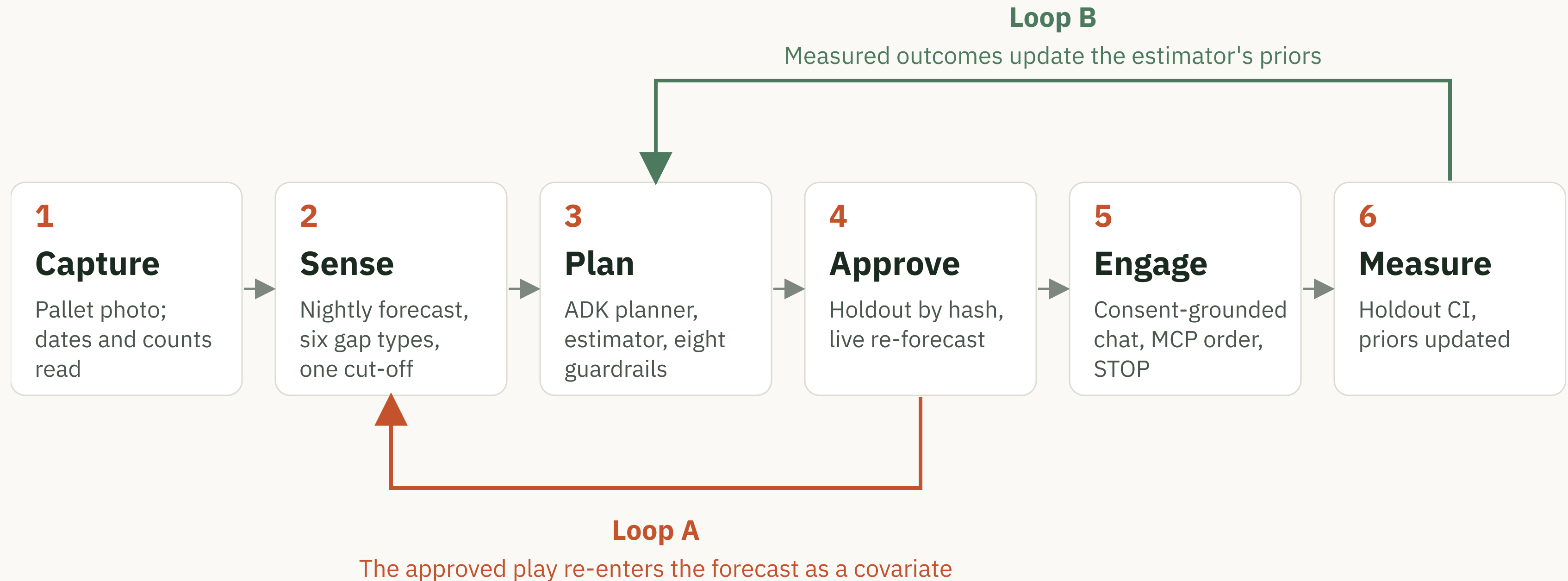


Measure

Treated vs holdout, 95% CI. Priors update from the counts.

PROCESS FLOW

Two closed loops, not an open-ended pipeline.



SCREENS

Guardrails reject; customers get Kannada plus English.

Recorded from Gemini · 28 Sept 2026

run_run_chips_ds07_v1_e9971541

Replay at 4x

Guardrail attempts · 4 proposed, 3 rejected

Attempt 1

REJECTED

+25880ms

cite_or_drop

uncited numbers in rationale: 52

▼ Rejected rationale

This play proposes a bundle of "Masala Chips 200G" with "Coconut Water 1L" at a price of 58.50 INR, offered through an outlet channel. This approach is preferred over a markdown for an online_sellby_breach gap, aligning with policy preference for bundles and transfers. The play targets all available segments (totaling 334 customers after consent) and is expected to clear 22.98 units, generating an expected margin of 191.88 INR. This play passes all guardrails, including maintaining a net margin of 14.27% which is above the 8.00% category floor for snacks. The alternative transfer play was rejected due to a negative expected margin of -161.52 INR.

Attempt 2

REJECTED

+40063ms

cite_or_drop

uncited numbers in rationale: 52

► Rejected rationale

Attempt 3

REJECTED

+55016ms

cite_or_drop

uncited numbers in rationale: 52

► Rejected rationale

Attempt 4

PASSED ALL GUARDRAILS

+63144ms

▼ Accepted rationale

This play proposes a bundle of "Masala Chips 200G" with "Coconut Water 1L" offered through an outlet channel. This approach is preferred over a markdown for an online_sellby_breach gap, aligning with policy preference for bundles and transfers. This play passes all guardrails. The alternative transfer play was rejected.

Chat as Meena

LIVE

2482 ms

CustomerMeena (DS-07, kn)▼

treated on the chips play once approved

ನಮಸ್ಕಾರ, ಇಂದು ಯಾವುದಾದರೂ ಆಫರ್ ಇದೆಯೇ?

ನಮಸ್ಕಾರ! ಹೌದು, ಮಸಾಲಾ ಚಿಪ್ಸ್ 200G ಬಂಡಲ್ ಮೇಲೆ ಆಫರ್ ಇದೆ. ಇದರ ಬೆಲೆ ₹58.5. 2026-10-15 ರೊಳಗೆ ಇದನ್ನು ಬಳಸಿದರೆ ಉತ್ತಮ. ಈಗಲೇ ಆರ್ಡರ್ ಮಾಡಲು 'Add to cart' ಬಟನ್ ಒತ್ತಿ.

Add to cart

Not now

STOP

English: Hello! Yes, there is an offer on Masala Chips 200G bundle. Its price is ₹58.5. It is best to use it by 2026-10-15. To order now, press the 'Add to cart' button.

Do you have Cola Zero?

Type a message...

Send

STOP

Right: customer chat, live

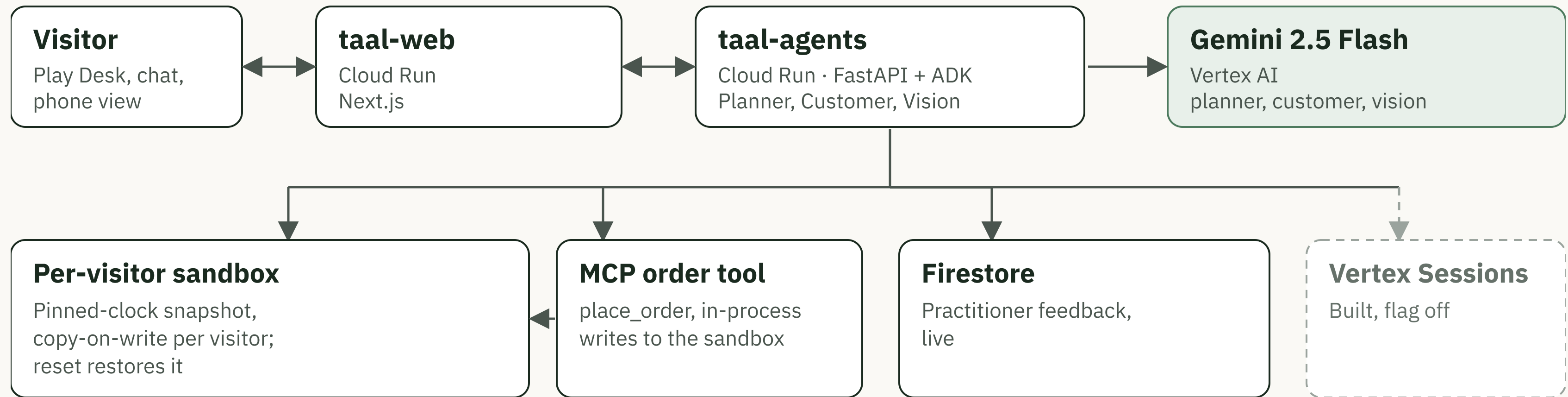
Kannada reply with an English gloss; the order goes through the MCP order tool.

Left: Play Desk trace, recorded run

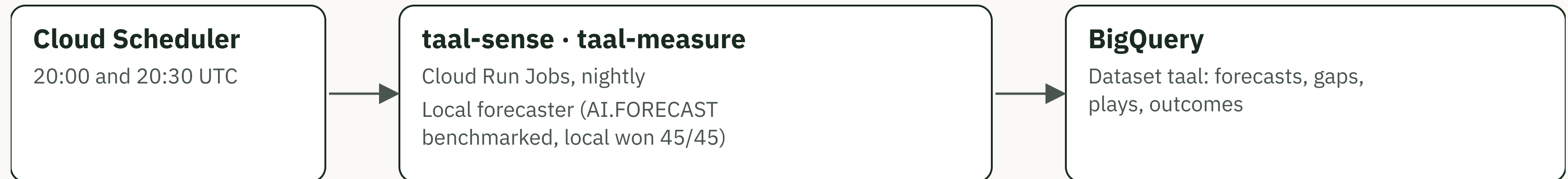
Three attempts rejected by cite-or-drop; the fourth passes all eight guardrails.

ARCHITECTURE

A sandbox serves judges; nightly jobs run on BigQuery.



NIGHTLY PATH



TECHNOLOGIES USED

Google Cloud AI stack, benchmarked before use.

Gemini 2.5 Flash

On Vertex AI: vision intake, Planner Agent and Customer Agent.

Google ADK

LoopAgent (max 3), 6 deterministic tools and 8 guardrails around every play.

BigQuery

Benchmarked AI.FORECAST and ARIMA_PLUS_XREG; the local forecaster is served.

Firestore

Practitioner feedback store, live in the deployed service.

Cloud Run

Services scale to zero; Cloud Run Jobs and Cloud Scheduler run Sense and Measure nightly.

Next.js • FastAPI • MCP

Play Desk, phone view and chat on Next.js; API on FastAPI; orders via an in-process MCP tool.

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

Priced from measured usage; sold on rupees rescued.

PER 30 DAYS, MODELLED FROM MEASURED USAGE

₹486 / \$5.86

Real Gemini tokens, 752 BigQuery jobs and Cloud Run time, priced at public list rates. Not a bill.

Method: modeled_from_measured_usage.

Forward-looking profiles ESTIMATES

Profile	Scale	Monthly
Micro	1 shop, ≤500 SKUs	₹200–500
Mid	300 SKUs, 10 nodes	₹2,500–5,000
Large	20,000 SKUs, 500 nodes	₹1–2 lakh

BUYER AND PRICING

Buyer: the category or supply-chain head, who owns write-off.

Basis: a fee per node per month, or a percentage of rupees rescued vs the holdout.

BREAK-EVEN ILLUSTRATION

≈ ₹21,710 – ₹30 per node

Per month: projected margin ₹347,356 and run cost ₹486, each over 16 nodes. Synthetic, not a quote.

Doubtful reads go to a person; outcomes say SYNTHETIC.

SKU GUESS	BEST BEFORE	FACINGS	CONFIDENCE
SKU-WHEAT-ATTA-1KG	2026-11-05	4	sku 80% · date 70% · count 90%
SKU-POHA-500G	2026-10-25	4	sku 75% · date 65% · count 85%

Is this Poha 500G, best before 25 Oct 2026, 4 facings?

Yes

No

Confirm rows

Phone view

A read under 70% confidence is asked back before it reaches inventory.

Measured

Holdout is a randomly assigned control group -- customers who see no offer for this play at all, chosen the same way as everyone in "Treated" except for a coin flip. Every lift number below is the difference between what the treated group actually did and what this control group did, not a before/after comparison against the same customers. This is the only honest way to know a play caused the change, rather than a trend that would have happened anyway.

0.4 kg

of food diverted from waste, across 2 measured plays

1.0 kg

CO2e avoided (estimate; see the emissions factor caveat in docs/DATA_MODEL.md)

₹50

margin preserved that would otherwise have been written off

Run Measure

2 plays joined against orders: 2 measured, 0 unmeasured (below min treated n) · 12 Sept, 09:00 am

PLAY	TREATED	HOLDOUT	LIFT	WASTE AVOIDED	CEO NUMBER	DATA	LOOKER
SKU-MASALA-CHIPS-200G DS-07 · bundle min treated n 20 · measured 12 Sept, 09:00 am	Treated 316 customers, 1 responders, 2 units margin ₹9, discount ₹7	Holdout 36 customers, 0 responders, 0 units margin ₹0, discount ₹0	0.3% (CI -9.3%–1.8%)	₹50 0.4 kg	1.31 net margin recovered per ₹1 of discount vs holdout	• SYNTHETIC	Looker (n/a)
SKU-KAJU-KATLI-250G DS-03 · preorder min treated n 20 · measured 12 Sept, 09:00 am	Treated 295 customers, 0 responders, 0 units margin ₹0, discount ₹0	Holdout 28 customers, 0 responders, 0 units margin ₹0, discount ₹0	0.0% (CI -12.1%–1.3%)	₹0 0.0 kg	unmeasured	• SYNTHETIC	Looker (n/a)

Outcomes: treated vs holdout, SYNTHETIC badge

316 treated, 36 holdout; the lift interval crosses zero and is reported as such.

PERFORMANCE REPORT

Measured, dated, and traced to a raw file.

PLANNER, LIVE28 Sep

5/5

proposed, 0 fallbacks; 34–65 s each

eval/raw/planner_real_traces_2026-09-28/summary.json

CUSTOMER AGENT28 Sep

4.56 s_{p50}

p95 6.48 s, 0.48 s over the 6 s budget; 50/50 returned 200

eval/raw/customer_latency_2026-09-28/analysis.json

APPROVE + RE-FORECAST21 and 28 Sep

8.3–10.6 s

two live measurements; write-off ₹9,194 → ₹8,068

eval/raw/sweep_vertex_2026-09-21.txt · eval/raw/post_deploy_check_2026-09-28/summary.json

AGENT SIMULATION21 Sep

190/190

live Vertex; stock, holdout arm, stacking, STOP, hallucination

eval/raw/agent_simulation_2026-09-21.json

VISION INTAKE21 Sep

30/30

correct dates; synthetic photos only

eval/raw/vision_synthetic_2026-09-21.json

BACKTEST, SYNTHETIC20 Sep

0.113 WAPE

MAPE 0.133 over 63 tier-origin rows

eval/raw/external_backtest_2026-10-09.json · eval/raw/backtest_rows_2026-09-20.jsonl

LIVE EVAL ON VERTEX2026-10-09

15/15

rationale judge; same-family judge, weak evidence

eval/raw/live_eval_2026-10-09/summary.json

38/46

ADK response match

6/46

exact trajectory; retry-order variance

Public non-grocery dataset backtest: WAPE 0.83 (UCI Online Retail II). Method and caveats in eval/external_backtest.md.

7 usable practitioner responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json)

3 of 7

write off stock weekly or daily

3 of 6

online sellers say the sell-by rule limits what they sell

0 of 7

measure a promotion against a control group

4 of 7

want a one-week pilot

"Blanket markdowns erode margin on products that may still sell normally."

Category / merchandising, quick-commerce

"Item-store recommendations with incremental-margin estimates."

Category / merchandising, quick-commerce

Anonymous by consent; contacts never published. Counts, not percentages.

Next: a real pilot, real photos, more tenants.

1

A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout.
Metric pre-registered in docs/pilot.md.

DESIGNED

2

Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 is on synthetic ones.

HARNESS BUILT

3

Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LINKS

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GITHUB REPOSITORY



<https://github.com/Nandish3010/probable-fortnight>

LIVE APP



<https://taal-web-2obkp776ca-el.a.run.app>

DEMO VIDEO

QR added with the video link

link added at submission